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ORIGINAL RESEARCH

Projecting climate change impacts on the distribution of the most captured tuna species on Atlantic waters

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Abstract
Tunas are among the most valuable marine resources and some of the most popular seafood worldwide, traded on a global scale. Not only do they support the livelihoods of fishers, they also play a crucial ecological role as both predators and prey. The challenges already faced by fisheries governance in managing these highly migratory species are likely to intensify due to climate-induced changes. This study investigates potential shifts in habitat suitability and geographical distribution of the most captured tuna species in the Atlantic Ocean by 2030 and 2050, across three Shared Socioeconomic Pathway (SSP) scenarios (i.e., SSP1-2.6, 4–6.0, and 5-8.5; CMIP6) using species distribution models (SDM) through MaxEnt. Its findings indicate the start of a potential poleward shift in tuna distribution as early as 2030, accompanied by significant declines in habitat suitability at lower latitudes. This trend poses a risk of range restriction or even local extinction of this group in equatorial regions. Given the socioeconomic importance of tuna species, predicting their responses to climate change could inform effective management strategies ensuring not only the sustainability of tuna populations but also the livelihoods of the communities that depend on them. Future research should incorporate the potential impact of changing environmental conditions on fishing intensity, as this will further shape the long-term distribution of these species.

Keywords Species distribution models · Fisheries · Sustainability · Atlantic

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Introduction

Tunas are among the most important fishing resources in the world in terms of landings and economic value (Xie et al. 2023) and also among the most popular seafood in the world so their trade has been developed on a global scale (Guillotreau et al. 2017). Indeed, they constitute one of the most valuable groups of marine species, with their catches increasing to 8.3 million tonnes in 2022—the highest level recorded (FAO, 2024). The main commercial tunas contributed with a total of 4.95 million tonnes of catch in 2021, 57% of the catch being skipjack tuna (*Katsuwonus pelamis*), followed by yellowfin (31%, *Thunnus albacares*), bigeye (7%, *T. obesus*) and albacore (4%, *T. alalunga*). Atlantic bluefin tuna (*T. thynnus*) accounted for just 1% of the global catch (FAO, 2024). The assessment and management of the major commercial oceanic tunas is made by five Tuna Regional Fishery Management Organizations (RFMOs) around the world. Atlantic tunas are managed by the International Commission for the Conservation of Atlantic Tunas (ICCAT), which regulates approximately 30 tuna and tuna-like species and is responsible for biometrics, ecology and oceanography research, focusing on the effects of fishing on these resources. ICCAT's assessments underpin the scientific advice for management that is provided to the Commission. Their main aim is to evaluate the sustainability of current and proposed future harvest practices in light of the Commission's objective to maintain the populations at a level that permits their maximum sustainable catch (ICCAT 2024a).

Currently, there are 53 Contracting Parties that capture tuna species in the Atlantic Ocean (ICCAT 2024b). Captures of Atlantic Ocean stocks represented approximately 11% of the world's production of tuna (ISSF 2021), with the five most captured tuna species (i.e., skipjack, yellowfin, bigeye, albacore and bluefin) (ICCAT 2024c) in 2019 constituting approximately 555 tons - a 7% decrease from 2018 catch levels. Overall, a general tendency for the total catch to decline since the mid-1990s, followed by a new upward trend since 2009 has been observed in this ocean (ISSF 2021). Contrarily, global tuna catches have been increasing, mainly due to the exploitation of Pacific and Indian Ocean stocks (FAO, 2024).

Its high demand raised several issues regarding stock status and management failures, making sustainable management critically important (McKinney et al. 2020). Moreover, tunas are an important commodity that supports the livelihood of many fishers, never forgetting the need to balance it with their ecological importance as predators and prey (McKinney et al. 2020). Tunas possess anatomical, biological and physiological traits that contribute to their ecological success, including fast growth rates, high reproductive potential and remarkable swimming abilities that enable long-distance migrations across ocean basins (McKinney et al. 2020). However, these traits are not homogeneous within the group, with existing species-specific differences (Brill and Hodbay 2017). These differences influence how tunas respond to environmental stressors, including those driven by climate change. Climate-induced stressors are expected to exacerbate the challenges already faced by marine species and fisheries governance in general, with tunas being no exception (McIlgorm et al. 2010). Given the connection of tuna populations to global trade networks–even through small-scale fisheries (Lam et al. 2020) - there is a growing need for adaptive fisheries management (Bell et al. 2020), integrated in the context of 'climate-smart' marine spatial planning (Frazão Santos et al. 2025). The effectiveness of such adaptation measures depends on a range of social, economic, political and cultural factors (Sumaila et al. 2011). Thereby, promoting anticipatory, collaborative, and long-term approaches to management can enhance the resilience of both marine ecosystems and the communities that depend on them (Sumaila et al. 2011; Bell et al. 2020; Lam et al. 2020). An important step towards sustainable fisheries is to generate knowledge and predictions on the potential changes in species distribution. It enables researchers and policymakers to consider different scenarios which might conditionate natural ecosystems in the future or even key economic activities. For this purpose, combining biogeographic knowledge and modelling frameworks in climate change impact studies enables an attempted prediction of the future impacts of changing environmental conditions on biodiversity and ecosystem health (Hannah et al. 2002). In this regard, species distribution modelling (SDM) has been increasingly developed over the past few decades (Zimmermann et al. 2010; Robinson et al. 2011). These models offer a good framework for predicting and projecting changes to species distribution ranges across space and time, and regarding large species assemblages (Elith et al. 2006). Briefly, these models take geo-referenced species occurrence data and a desired set of predictors (which can be environmental, biological, anthropogenic, etc.) across a defined geographical extent. After choosing the desired algorithm (or an ensemble of different algorithms), the models establish the relationship between species occurrence and the predictor values present at those coordinates. This will define the species' ecological niche and enable its projection towards different arrays of environments across time and/or space (Miller 2010).

Erauskin-Extramiana et al. (2019) created species distribution models for tuna species to investigate the global effect of environmental conditions on the distribution and relative abundance of six tuna species and estimated the expected end-of-the-century changes. Temperate tunas (which included albacore and Atlantic bluefin) and the tropical bigeye tuna were expected to decline in the tropics and shift poleward. In contrast, skipjack and yellowfin tuna were projected to become more abundant in tropical areas as well as in most coastal countries' Exclusive Economic Zones (EEZ). The study has also demonstrated that tuna habitat has shifted poleward over the 1958–2004 period (Erauskin-Extramiana et al. 2019). Still, on a global scale, it was projected that the habitat of tropical tunas continues shifting poleward in response to climate change and ocean warming (Monllor-Hurtado et al. 2017). Some studies have shed light on the relationship between some of the most important tuna species distribution and climate change namely at a regional level, projecting an expansion in area and a shift of productive areas (Vaihola and Kininmonth 2023), and stating that ocean warming is expected to make higher latitudes more suitable habitats for tuna species that already dominate the area in terms of catch and effort (Mediodia et al. 2020). Other research endeaavours have also delved in this topic, through the use of mechanistic models (opposed to SDMs which are correlative models). Lehodey et al. (2013) evaluated climate-change impacts on the Pacific Skipjack tuna under different fishing effort scenarios, and Dueri et al. (2013) focused on evaluating potential impacts on habitat suitability at a global scale. Both studies employed numerical models developed to represent the impacts of environmental and anthropogenical predictors on skipjack population dynamics– the former using SEAPODYM, and the latter relying on the APECOSM-E Model. Most of these studies suggested and warned of the need to plan future national actions towards the proper management of these species.

The present study aims to project the potential changes to the habitat suitability, geographic distribution, and resulting species overlap of the main Atlantic tuna species (i.e., two temperate species: *Thunnus alalunga* and *T. thynnus*; and three tropical species: *Katsuwonus pelamis*, *T. obesus*, and *T. albacares*) across two future time frames (i.e., 2030 and 2050), and 3 Shared Socioeconomic Pathways (SSP) (i.e., SSP1-2.6, 4–6.0, and 5-8.5; CMIP6)– through the use of the most recent Coupled Model Intercomparison Project– i.e., CMIP6, from the AR6 (IPCC 2023) - using temperature, salinity, and oxygen content as environmental predictors, which have been shown to correlate with the abundance and distribution of tunas (Block and Stevens 2001).

Obtaining such projections is key in adequately informing the stakeholders responsible for the management and exploration of the Atlantic stocks of these species, with the ultimate goal of ensuring their long-term sustainability and the conservation of the marine ecosystems and livelihoods which depend on these species.

Materials and methods

The procedures described below were conducted using the open-source R statistical software (v.4.1.2; R Core Team 2021). All scripts, outputs and datasets are supplied in the supplementary materials.

Occurrence collection and curation

Geo-referenced occurrences for each of the five tuna species included in this study were obtained from both the Global Biodiversity Information Facility (GBIF, https://www.gbif.org/; GBIF 2024a-e) and the International Commission for the Conservation of Atlantic Tunas (ICCAT, https://www.iccat.int/). The data available in these sources refers mostly to fishery data. Indeed, most of the occurrences retrieved from GBIF originate from the NOAA Southeast Fisheries Science Center (SEFSC), Fisheries Log Book (FLS) System (United States Geological Survey 2024), the POPA Fisheries Observer Program of the Azores (Machete 2014), and the Fishes of Texas Project (FoTX) Database from Darwin Core (Hendrickson and Cohen 2024). At the same time, the ICCAT holds large datasets of several Atlantic-wide oceanographic surveys with a focus on tuna species, albeit also hosting data associated with fishery operations. The data retrieved was selected from worldwide occurrences to feature only those occurring within the Atlantic Ocean (i.e., between 60ºN and 60ºS) and the Mediterranean Sea. Restricting the data to these ocean basins enabled the restriction of model calibration only to the areas which are accessible to the populations of each species being analyzed (Peterson and Soberón 2012). This was conducted even though some of the species included feature a trans-oceanic distribution, since it is expected that the tunas in different ocean basins belong to separate populations, as shown by genetic differentiation in bigeye tuna across the Atlantic, Indian, and Pacific Oceans (Alvarado Bremer et al. 1998). Data from GBIF was limited to valid occurrences using a specific set of filters, specifically 'human observation', 'occurrence between 1970 and 2024', absence of geospatial issues, 'present' occurrence status, and a clear taxonomic ID for each species. The occurrence data was subject to a curation procedure to ensure the inclusion of accurate data and reduce noise, thus excluding data geo-located on land, duplicate observations (i.e., featuring the same coordinates), missing (NA) values in either latitude or longitude, and improper datum conversion points. To remove data occurring on land, the data was clipped using a shapefile of the world's ocean bodies (found on Natural Earth Data, https://www.naturalearthdata.com/*)*. All data was then further restricted in time to exclude points registered before 1992.

Environmental predictors

The environmental predictors were retrieved using the '*biooracler'* package (Assis et al. 2024), which allows for download and integration of the environmental layers supplied by Bio-Oracle. Bio-Oracle is an online repository which offers global geophysical, biotic, and climate layers at a common spatial resolution (i.e., 5 arcmins) and a uniform landmask (Assis et al. 2018, 2024). This repository features a set of present-day and projected future layers across different decades (between 2000 and 2100) and the different Shared Socio-economic Pathway scenarios (SSP) of the Coupled Model Intercomparison Project Phase 6 (CMIP6; O'Neill et al. 2016; Riahi et al. 2017). For the present study, three specific SSPs were chosen, namely SSP1-2.6 - the low-end range of future forcing pathways, simulates an optimistic scenario compatible with the 2ºC target, producing a forcing of 3.8–4.2ºC increase in atmospheric temperatures; SSP4-6.0– a middle-emissions scenario, forcing a warming of 3.5–3.8ºC.; and SSP5-8.5 - the high end of the range of future pathways, representing a high-end emission scenario capable of producing a radiative forcing of 8.5 W m-2 by the end of the century, capable of resulting in warming of 4.7–5.1ºC (Meinshausen et al. 2020). From these three scenarios, the maximum, mean, minimum, and range of sea surface salinity, sea surface temperature, primary productivity (i.e., phytoplankton), dissolved oxygen, and slope were initially retrieved, for three different time periods– specifically present-time (i.e., 2010–2020), and two future times: 2020-30 and 2040-50. For all predictors considered, only surface-level layers were included in the present study. To detect and minimize multicollinearity (i.e., a strong correlation) between two or more predictor variables, the function 'vif' from package 'car' was used (Fox and Weisberg 2019), testing the variables against each other to calculate variance-inflation. Variables featuring a VIF (variance inflation factor) greater than 10 were signaled for evaluation (Naimi et al. 2014), and predictor removal was conducted informed by both the VIF value and expert knowledge on the variable's ecological relevance. Prior to modeling, each predictor variable was reprojected to the cylindrical equal-area projection – *"+proj = cea + lat_ts = 0 + lon_0 = 0 + x_0 = 0 + y_0 = 0 + datum = WGS84 + no_defs")* to ensure the equal area of all cells across the extent of each predictor, a requisite of the MaxEnt algorithm chosen for the distribution models (Elith et al. 2010).

MaxEnt modelling

The use of MaxEnt as the only modelling algorithm in the present study is supported by the consistent high accuracy demonstrated by this algorithm, and the common outperforming of other SDM approaches (Phillips and Dudík 2008; Feng et al. 2019; Shipley et al. 2022). While ensembling techniques featuring multiple algorithms are on the rise, the use of MaxEnt alone allows for the limitation of the introduction of different types of error and uncertainty into the final model (Elith et al. 2010). The SDM analysis followed the workflow described by Shipley et al. (2022) in their 'megaSDM' package, which is based on the MaxEnt algorithm, and is presented in the supplementary code. Briefly, the package takes specific raster stacks (for the present-day and for each of the future time periods and/or scenarios), manipulating and standardizing all predictor variables (either through re-projection, clipping, or re-sampling when necessary). In this process, the training (i.e., where occurrence and background points are located), and the study (i.e., where the model will be projected) areas are defined. The occurrence data is then manipulated and curated to mitigate the bias typically associated with collected or downloaded data and which can hinder the accuracy of the SDM predictions (Phillips et al. 2009; Beck et al. 2013; Varela et al. 2014).

This package then applies a procedure of environmental filtering of the occurrence data, with the goal of removing clustered or oversampled records, keeping the range of environments in which a species is found (Varela et al. 2014). This is achieved by dividing the environmental values at each point into an a priori set of bins (which in the present study featured an $n=25$). Then, one point from each unique bin combination is selected, resulting in a subset of occurrence points filtered by the environment (Varela et al. 2014; Castellanos et al. 2019; Shipley et al. 2022). In Table 1, the pre- and post-curation number of occurrence records is supplied for each species.

Background points—i.e., artificially created species occurrence points describing the environmental conditions of the training area - were generated for each species of tuna, enabling the integration of pseudo-absence data into the models (Peterson and Soberón 2012). This was done using a combined method, which employed both random and spatially constrained sampling—with a relative weight of 50% each, for each species (Shipley et al. 2022). The choice of a combined method reduces the likelihood of model overestimation, typically associated with regions of greater occurrence densities (i.e., spatial bias of relatively better-sampled regions) and which the random generation alone does not address (Lobo et al. 2010; Kramer-Schadt et al. 2013). Concurrently, this choice also reduces the likelihood of extreme extrapolation errors and overfitting which are a consequence of spatially constrained background-generation methods (Radosavljevic and Anderson 2014). A process of environmental filtering (Varela et al. 2014) was also applied to the resulting background points datasets, thus ensuring an even spread across available environmental space, at the same time retaining the desired spatial weighting (Shipley et al. 2022).

In total, half of the background points per species were randomly sampled from the entire study area (Barbet-Massin et al. 2012), with the remaining half being sampled from within a buffered region around each true-occurrence point - the radius of which being proportional to the 95% quantile of the distance to the nearest neighbor.

Each species' present-day habitat suitability was estimated using the MaxEnt modelling algorithm, which employs maximum entropy methods and is resilient to the use of presence-only species records (Elith et al. 2010), while at the same time featuring a highly competitive predictive performance when compared with other high-performing methods (Elith et al. 2006; Feng et al. 2019). For this, the 'megaSDM' package employs MaxEnt modelling using replication and subsequent ensembling of model predictions. For the present study, a replicate number of 5 model runs was employed for each species, each utilizing a different randomized subset of occurrence and background points (Shipley et al. 2022). Prior to ensembling, the area under the curve (AUC) was used as a metric to evaluate each model replicate, comparing the AUC of each replicate with the AUC of a null model (Raes and ter

Table 1 Species occurrence sample size before and after curation and environmental filtering. Post-curation data refers to the data directly used by the 'megasdm' package for maxent modeling

Habitat	Species	Pre-curation	Post-curation
Temperate	*Thunnus alalunga*	6084	1287
	Thunnus thynnus	7547	583
Tropical	*Katsuwonus pelamis*	16,495	651
	Thunnus albacares	14,167	1588
	Thunnus obesus	11,224	752

Steege 2007; Shipley et al. 2022). The final ensemble model was also evaluated in this way, with all models exhibiting an AUC of under 0.70 being removed.

The models were then projected onto the specified future time periods (i.e., 2030 and 2050) across all SSP scenarios (i.e., SSP-119, 126, 460, and 585), with the resulting outputs being ensemble maps of the median value of each pixel, thus reducing the effect of potential outliers (Araújo and New 2007; Shipley et al. 2022). These ensemble maps were used for obtaining the desired outputs following the post-analysis procedure described in the next section. The model evaluation plots and tables for each species models are supplied in the supplementary materials.

Post-analysis

Latitudinal distribution

Changes in habitat suitability and distribution along the latitudinal profile were assessed for each species. First, the latitudinal centroid was obtained for each time frame and climate scenario. The centroids were produced by calculating the arithmetic mean latitude of the cells occupied by the species (akin to Borges et al. 2022). Afterwards, the trends in mean habitat suitability along the latitudinal axis were determined, by converting each ensemble raster into a matrix and calculating the mean value for each row. The values from the resulting present-day vector were subtracted from those of the 2030 and 2050 vectors, for each SSP the resulting output plotted.

Projected tuna distribution overlap

To evaluate potential changes to tuna biodiversity in the study region, a threshold value was applied to the habitat suitability ensembles, to generate binary maps of probability of occurrence (i.e., presence/absence or 0/1). Briefly, the 'maximum test sensitivity and specificity' logistic value was chosen for this, as it has been shown to be quite effective when using presence-only data, generating models which consistently outperform those using other types of threshold (Liu et al. 2016; Shipley et al. 2022). For each time frame and each climate scenario, these binary maps were overlapped and their same-cell values were summed to obtain a map with the predicted present-day and projected future tuna richness, and evaluate potential change.

Results

Contribution of environmental predictors

The environmental predictors influencing the habitat suitability and consequent occurrence of each included tuna species varied (Table 2). For *T. albacares* and *T. thynnus*, the most contributing variable (relative to those included) was dissolved O_2 minimum (i.e., 41.9% and 22.6% respectively). For *T. alalunga* and *T. obesus*, the most contributing predictors were layers related to salinity (i.e., maximum, with 22%, and mean, with 17.8%, respectively). Lastly, the maximum temperature layer was the most contributing variable for *K.*

Table 2 Relative contribution of the environmental predictors to the maxent model. Only the top 4 predictors are presented for each species. Further details are presented in the supplementary materials

Habitat preference	Species	1st	2nd	3rd	4th
Temperate	*Thunnus alalunga*	Salinity maximum (22%)	Temperature maximum (18.1%)	Total phytoplankton minimum (13.9%)	Dissolved oxygen minimum (10.1%)
	Thunnus thynnus	Dissolved oxygen minimum (22.6%)	Temperature maximum (20.8%)	Temperature mean (19.1%)	Dissolved oxygen maximum (17.8%)
Tropical	*Katsuwonus pelamis*	Temperature maximum (35.2%)	Dissolved oxygen minimum (26.5%)	Total phytoplankton maximum (8.1%)	Salinity maximum (7.9%)
	Thunnus albacares	Dissolved oxygen minimum (41.9%)	Temperature maximum (18.6%)	Salinity maximum (9.8%)	Dissolved oxygen maximum (8.8%)
	Thunnus obesus	Salinity mean (17.8%)	Temperature maximum (15.5%)	Salinity maximum (15.5%)	Total phytoplankton minimum 15.5%)

pelamis (35.2%). However, temperature maximum was the 2nd most contributing variable for all species (apart from *K. pelamis*, for which minimum dissolved O_2 minimum featured as the 2nd most contributing variable, with 41.9%). For all species, slope was either the least contributing variable (i.e., 4.5% for *T. alalunga* and 0.7% for *T. thynnus*) or the second least contributing variable (≤2% for the remaining species).

General patterns of habitat suitability

Figures 1 and 2 present the resulting maps of habitat suitability for all species considered, for the present-day baseline (i.e., 2010) and the projected suitability for the furthest time period considered (i.e., 2050) and SSP5-8.5. Over time, all species included in the present study exhibited a shift of their latitudinal distribution centroid, compared to their respective predicted present-day baselines (Fig. 3). For all species, the centroids are projected to be moving southward over time (between the present-day baseline, 2030, and 2050), except for *T. thynnus*, which features very similar values for both 2030 and 2050, regardless of the climate scenario (Fig. 3B). These trends are not only exacerbated with time, but also with the pathway considered: specifically, SSP4-6.0 in the year 2050 is the scenario where most species undergo their furthest southward movement (apart from *T. thynnus* and *T. albacares*).

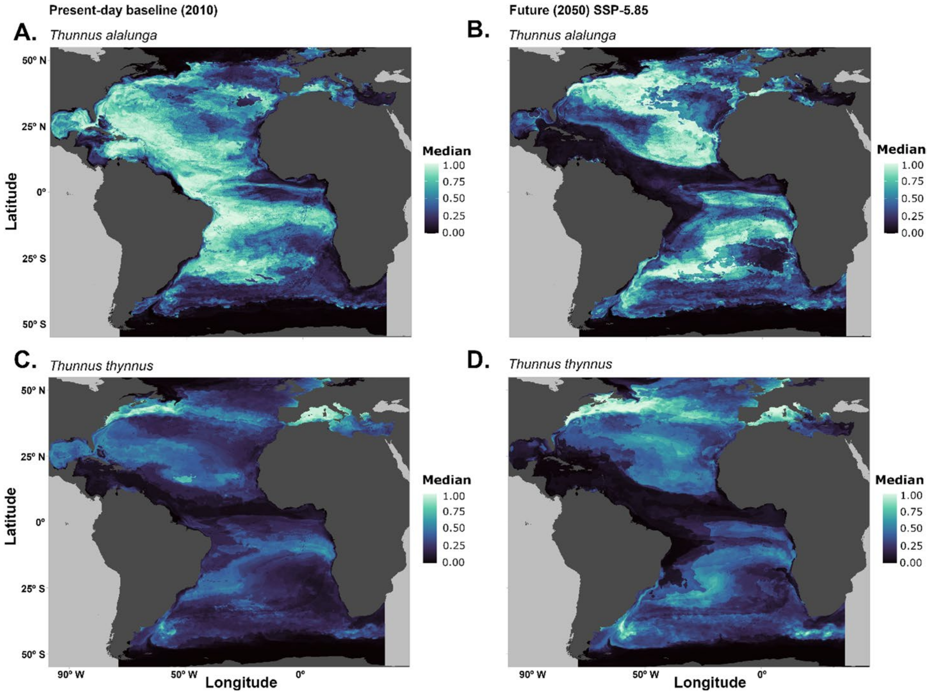

Fig. 1 Spatial projection of the predicted habitat suitability maps of the temperate tuna species [i.e., *Thunnus alalunga*(**A**, **B)** and *T. thynnus*(**C**, **D)**] in the Atlantic Ocean, for the present-day (**A and C**) and the future time for SSP5-8.5 (**B** and **D**). The maps for the remaining time periods and SSP scenarios considered are supplied in the supplementary materials

Trends in habitat suitability

Regarding the projected trends of change in mean habitat suitability over time, along the latitudinal profile, there is an apparent effect of time and pathway scenario for all species, with most exhibiting clear decreases in mean habitat suitability along the most equatorial latitudes, compared to the predicted present-day baseline (Figs. 4 and 5). These decreases are, in general, followed by relative increases in the suitability of habitat at the northernmost and southernmost latitudes. However, each species' response differs. Regarding the temperate species, *T. alalunga* is projected to undergo its greatest decrease between 10ºN and 10ºS (up to −0.38%), regardless of the pathway scenario (Fig. 4A–C). Overall, this species exhibits decreases in habitat suitability along most latitudes above 10ºN - i.e., mostly in the northern hemisphere - and shows the largest positive change above 40ºN (up to +0.25% under SSP1-2.6, Fig. 4A). Increasing habitat suitability is also observed between 15ºS and 45ºS, albeit with a smaller difference relative to the present-day baseline (at a highest value of 0.13% under SSP4-6.0, Fig. 4B). For this species, SSP4-6.0 hosts the greatest relative change in mean habitat suitability compared to the present-time (Fig. 4B). Concerning *T. thynnus* (Fig. 4D–F), it exhibits a comparatively smaller negative change in mean habitat suitability, achieving just over—0.13% under SSP5-8.5 at 10ºS (Fig. 4F). In this species, the greatest decreases occur at approximately 10ºS and 25ºN for all SSP scenarios, while a

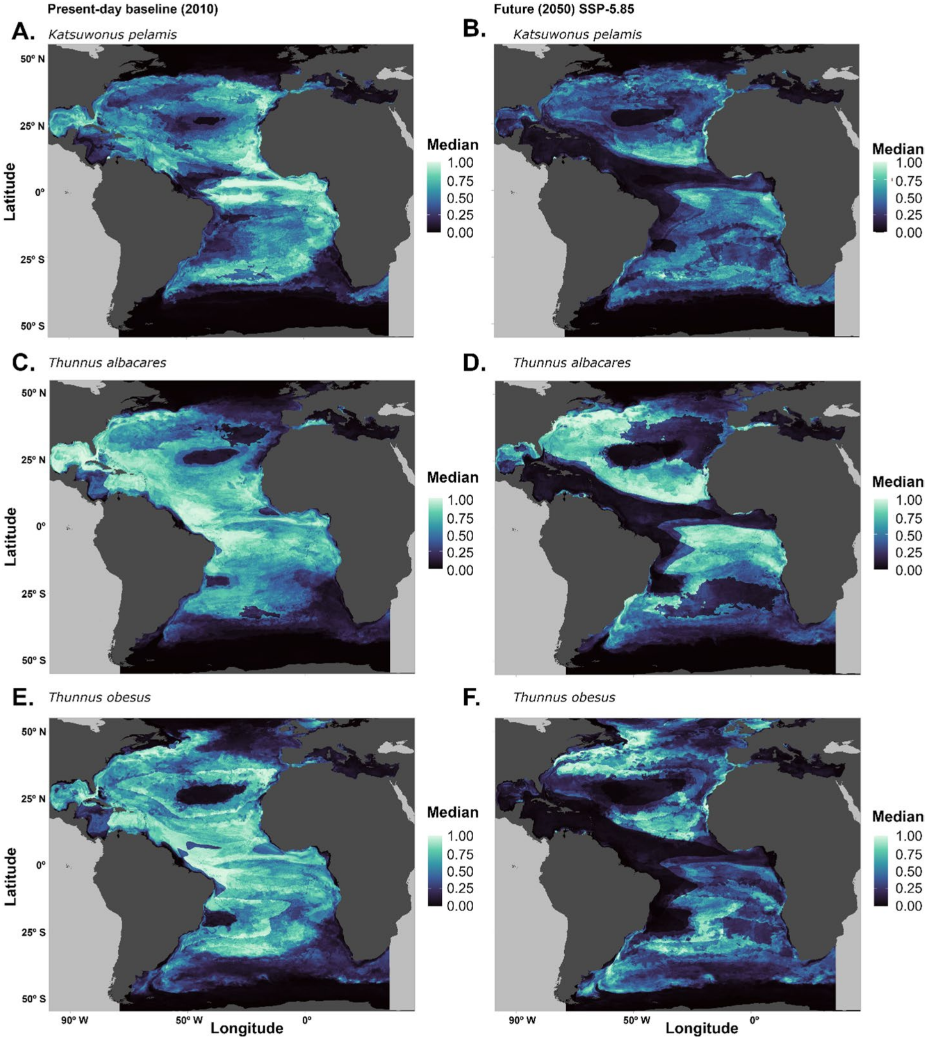

Fig. 2 Spatial projection of the predicted habitat suitability maps for the tropical tuna species *Katsuwonus pelamis* (**A**, **B**), *Thunnus albacares* (**C**, **D**), and *Thunnus obesus* (**E**, **F**) in the Atlantic Ocean, for the present-day (**A**, **C** and **E**) and the future time for SSP5-8.5 (**B**, **D** and **F**). The maps for the remaining time periods and SSP scenarios considered are supplied in the supplementary materials

considerable positive change is observed for all pathways above 30ºN (up to +0.25% for SSP1-2.6 and over for SSP4-6.0 and 5-8.5), peaking at approximately 40ºN. In the southern hemisphere, a consistent increase in habitat suitability is observed below 10ºS, just under +0.13% for all scenarios.

With regards to the tropical species (Fig. 5), there is a clear pattern of decreasing habitat suitability which is exacerbated as time advances, worsening alongside the increasing severity of the SSP scenario considered. Indeed, SSP4-6.0 and 5-8.5 are the pathways where the three species considered undergo the greatest negative changes. Specifically, *K. pelamis* exhibits negative changes in habitat suitability along most of the latitudinal pro-

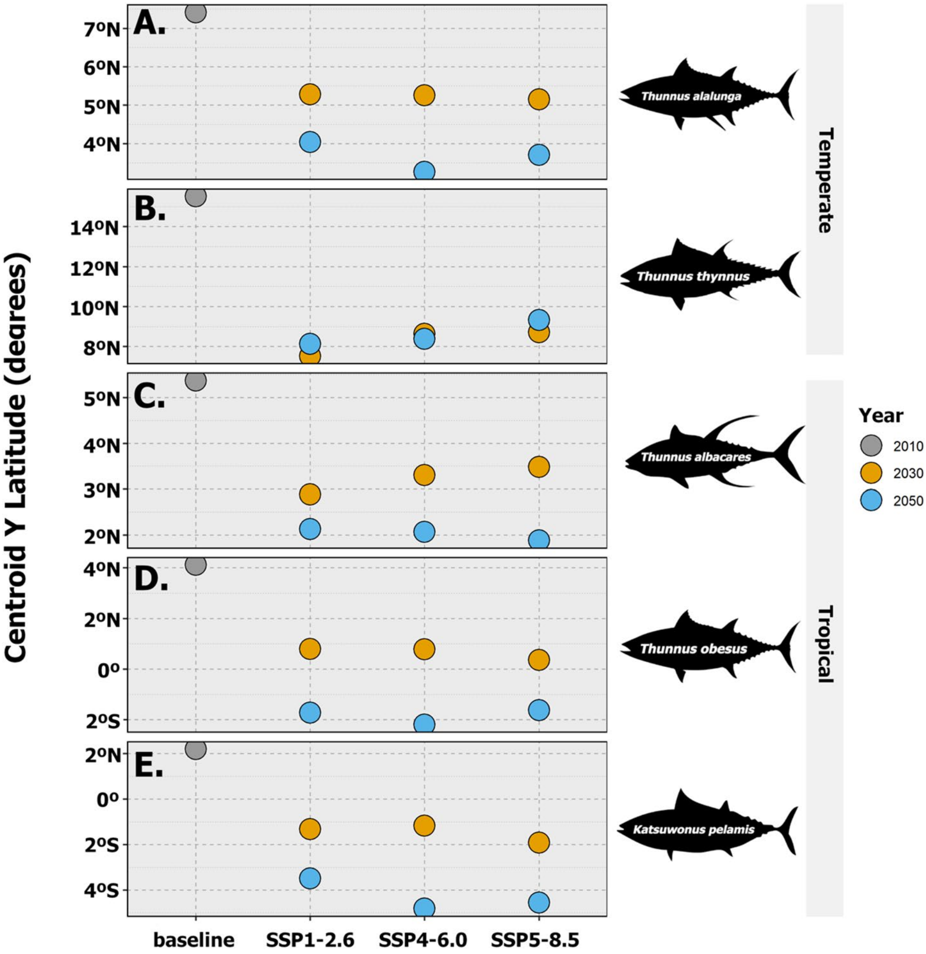

Fig. 3 Projected changes to temperate [**(A)**—*Thunnus alalunga*; **(B)**—*T. thynnus*] and tropical [**(C)**—*T. albacares*; **(D)**—*T. obesus*; and **(E)**—*Katsuwonus pelamis*] Atlantic Ocean tuna species' latitudinal distribution centroid (i.e., mean latitude of the occupied pixels) across the different Shared Socioeconomic Pathways (SSP, SSP1-2.6, 4–6.0, and 5-8.5). Each circle represents either the present-day (i.e., grey circles) or one of the future time frames (i.e., gold circles for 2030, blue circles for 2050)

file between 35ºS and 40ºN, with a maximum change of approximately −0.40% near the equator (Fig. 5A–C). Small latitude ranges exhibit limited changes towards gains in habitat suitability, namely between 35–45ºS and 40–50ºN up until 2050. *T. albacares* habitat suitability decreases relative to the present-day baseline between 30ºS and 30ºN, with the highest difference occurring between 0º and 10ºN for all SSP scenarios, at a value of approximately—0.40% for SSP1-2.6 and 4–6.0 (Fig. 5D and E) and just under −0.5% for SSP5-8.5 (Fig. 5F). It increases between 30–45ºS and 30–55ºN and with the strongest increase under SSP4-6.0 (i.e., a peak just under +0.25% at 45ºN; Fig. 5E). Lastly, *T. obesus* exhibits the greatest negative trend in habitat suitability (Fig. 5G-I), achieving values of over −0.25% between 10ºS and 25ºN (peaking at −0.5% for both SSP4-6.0 and SSP5-8.5; Fig. 5H and I). This species exhibits increases in habitat suitability at the highest latitudes, namely peak-

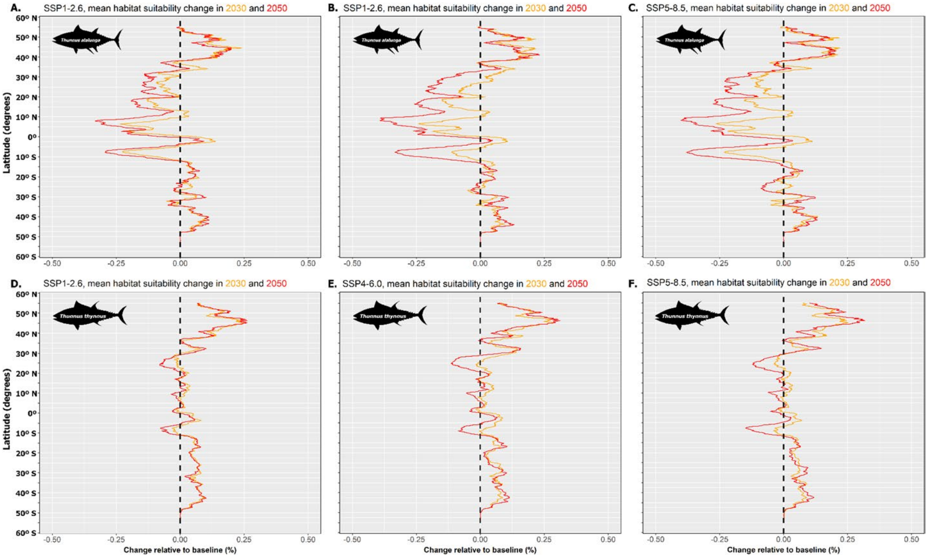

Fig. 4 Projected changes (%) in the mean latitudinal habitat suitability of the temperate tuna species [i.e., *Thunnus alalunga***(A–C)** and *T. thynnus***(D–F)**] in the Atlantic Ocean, across the different Shared Socio-economic Pathways considered [i.e., SSP1-2.6 **(A** and **D)**, SSP4-6.0 **(B** and **E)** and SSP5-8.5 **(C** and **F)**, for the years 2030 (orange lines) and 2050 (red lines). Changes were calculated relative to the predicted present-day baseline (represented by the dashed vertical line in each plot). Data to the left of the dashed line represents a relative decrease in mean habitat suitability, while data to the right represents a relative increase

ing at +0.13% between 30–50ºS for all SSPs in the southern hemisphere, and exhibiting a considerable increase between 40–55ºN, peaking at +0.25%.

Overall, both temperate and tropical species exhibit sparse oscillations towards gains by 2030 at lower latitudes (Figs. 4 and 5), albeit for tropical species these are followed by a sharp decrease until 2050 (Fig. 5).

Regarding relative species overlap (concerning only the species included in the present study), the overlap of all species' binary maps revealed relatively high values for most of the Atlantic Ocean in the present-day, with exceptions being the areas with higher latitudes (near the Southern Atlantic Ocean and the Arctic Ocean), and within the deep Mediterranean (i.e., towards the Central and Eastern areas). Most of the modeling extent reveals overlap values of 3 or more species, revealing the extensive distribution of these species (Fig. 6).

Over time, species overlap undergoes considerable changes along the Atlantic, with an expansion of the areas with higher values (i.e., more species) towards higher latitudes, and a severe restriction in areas near the equator and the tropics (Fig. 7). This effect is present across the different timeframes and scenarios but is exacerbated over time and with the severity of the SSP considered. Specifically, SSP4-6.0 and 5-8.5 are expected to lead to a severe decrease in tuna species overlap along the American coastline (until 25ºS) and the Gulf of Mexico (Fig. 7D and F). Likewise, the equatorial areas of the African coastline are also projected to greatly reduce their species overlap. This overlap of distributions is projected to increase northward of the Bay of Biscay, with the expansion of several species

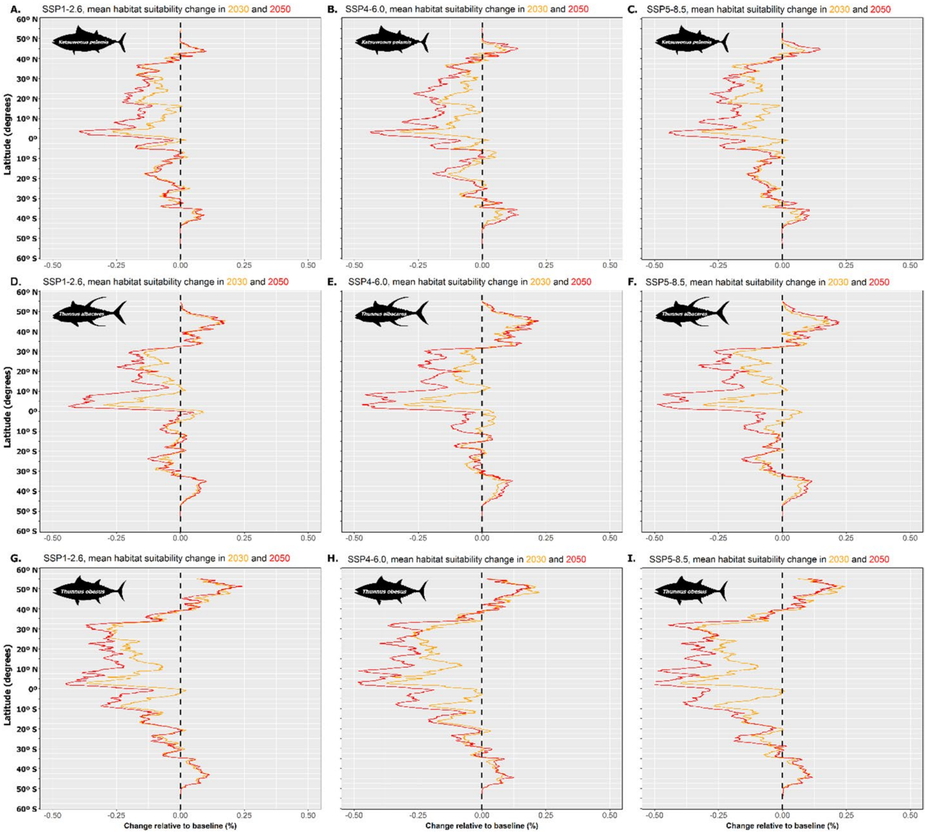

Fig. 5 Projected changes (%) in the mean latitudinal habitat suitability of the tropical tuna species [i.e., *Katsuwonus pelamis***(A-C)**, *Thunnus albacares***(D-F)**, and *T. obesus***(G-I)**] in the Atlantic Ocean, across the different Shared Socioeconomic Pathways considered [i.e., SSP1-2.6 **(A**, **D**, and **G)**, SSP4-6.0 **(B**, **E**, and **H)** and SSP5-8.5 **(C**, **F**, and **I)**, for the years 2030 (orange lines) and 2050 (red lines). Changes were calculated relative to the predicted present-day baseline (represented by the dashed vertical line in each plot). Data to the left of the dashed line represents a relative decrease in mean habitat suitability, while data to the right represents a relative increase

towards higher latitudes across all scenarios (Fig. 7). Such expansion of higher overlap values is also seen below 25ºS along both the South American and southern African coastlines.

Discussion

Tunas are considered apex predators, with a key role in top-down control of open water ecosystems (Juan-Jordá et al. 2011). Tunas also became one of the world's most important groups of exploited fish species, harvested to feed global supply chains. Therefore, fishing fleets and tuna markets have been affected by economic globalization (Mullon et al. 2017) and climate change is an additional threat that the species and their respective fisheries currently face, besides social and economic changes (Rubio et al. 2024). The emergence of climate change represents one of the most direct links between the environmental and

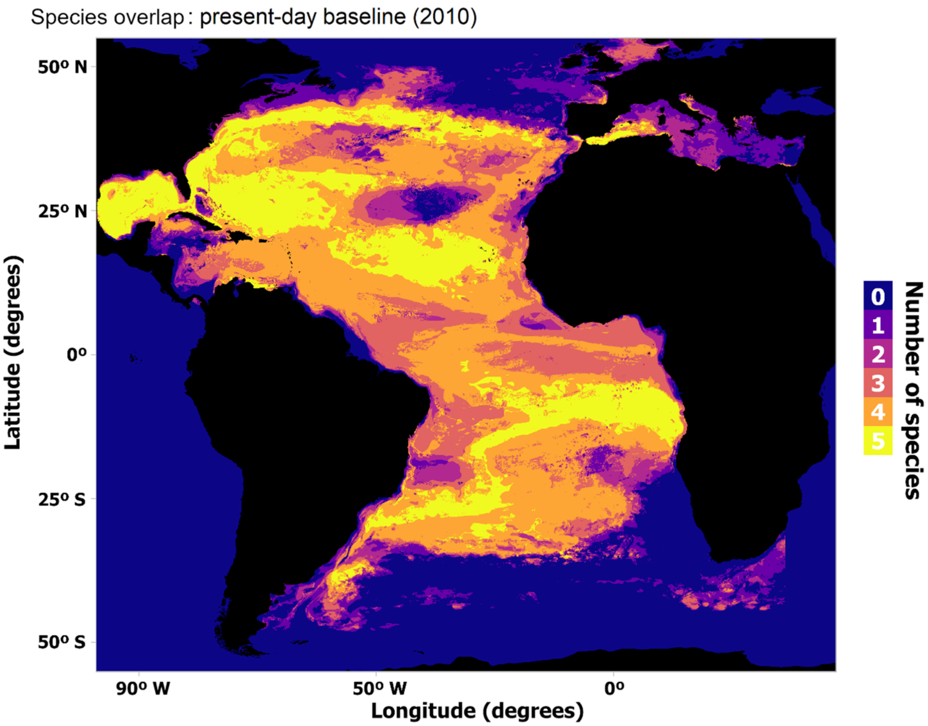

Fig. 6 Predicted present-day spatial species overlap (regarding the species included in the study, i.e., *Thunnus alalunga*, *T. albacares*, *T. obesus*, *T. thynnus*, and *Katsuwonus pelamis*) in the Atlantic Ocean. This map was obtained by overlap of each species' predicted presence-absence binary map

social dimensions of fisheries (Sumaila et al. 2016). Overall, the present study's results hint at potential macro-ecological and economic impacts of marine climate change, which will severely condition the distribution of these five tuna species, with potential cascading consequences on ecosystems and ecosystem services.

Indeed, the present study suggests a potential shift in tuna distribution towards higher latitudes (also known as poleward distribution shifts; Hastings et al. 2020) as soon as 2030, with severe decreases in habitat suitability which present a risk for restriction or even local extinction in areas near the equator (e.g., along the Gulf of Mexico, the Central/South American Atlantic coastlines, and the Gulf of Guinea). Habitat suitability was considerably impacted by the projected mid-century trends of temperature, dissolved oxygen, and salinity, for all species, and substantial change regarding present-day values was observed for all but one species (i.e., *T. thynnus*) along lower latitudes (although for *T. thynnus* a decrease in habitat suitability is also projected, albeit at a smaller magnitude). Sea surface temperature and surface salinity are known to be of particular importance for immature tunas, while temperature and oxygen content are highly relevant for adults, namely at the time of spawning (Chen et al. 2005). The fact that the marine productivity predictor (i.e., total phytoplankton) did not feature high on the most contributing predictors for most species - being the 3rd most contributing for *T. alalunga* and *K. pelamis*, and the 4th for *T. obesus* - could be explained by the time-lag between the peaks in the abundance of primary producers and the

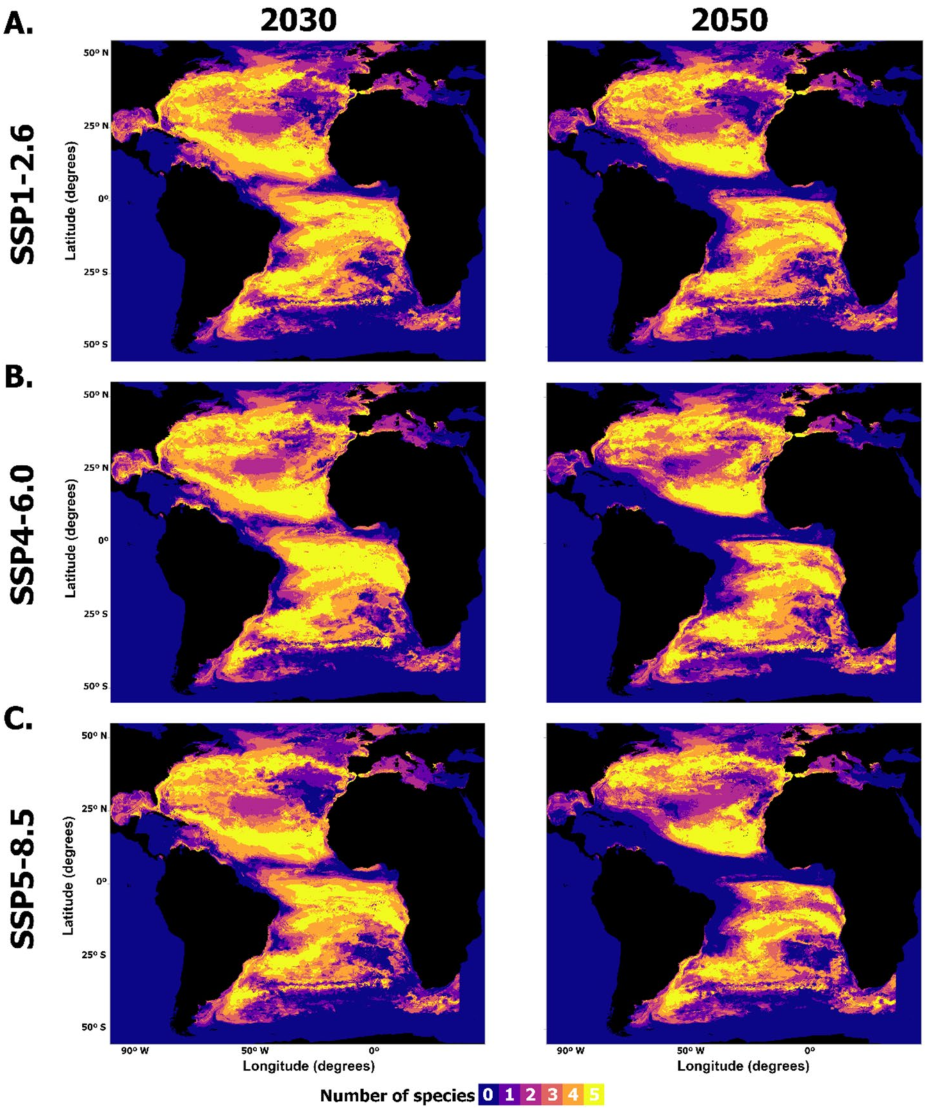

Fig. 7 Projected future-time spatial species overlap (regarding the species included in the study, i.e., *Thunnus alalunga*, *T. albacares*, *T. obesus*, *T. thynnus*, and *Katsuwonus pelamis*) for SSP1-2.6 (**A**), SSP4-6.0 (**B**), and SSP5-8.5 (**C**), for 2030 (left) and 2050 (right), in the Atlantic Ocean. These maps were obtained by overlap of each species' predicted presence-absence binary map

subsequent peak in the species on which tunas feed. Nonetheless, total productivity was able to explain approximately 10% of the data for three species.

This decrease in suitability along tropical areas with a concurrent (but not compensatory) increase over higher latitudes has previously been described (Barange et al. 2014; Dueri et al. 2014). Similar patterns have been shown in previous modelling approaches, where at a global scale tuna species can be observed to shift their distribution poleward when faced with climate forcing (Erauskin-Extramiana et al. 2019). The Atlantic populations of the tuna species considered in this study feature spawning areas in the Atlantic Ocean which are located in the regions projected to undergo the greatest decreases in habitat suitability. Indeed, these species are known to spawn in the regions close to the equator. For instance, *T. alalunga* typically spawns in tropical and subtropical waters approximately 10–25º from the equator, while *K. pelamis* spawns over a wide equatorial region between the Gulf of Guinea and the Gulf of Mexico (Andrade and Santos 2004; Benevenuti et al. 2019). This is also the case for *T. albacares* and *T. obesus*, which spawn year-round over a wide area of the tropical and subtropical Atlantic Ocean (ICCAT 2019). Recruitment levels are highly influenced by environmental conditions, namely in tropical tunas, suggesting a negative influence on projected future temperature levels in these species. While these species are regionally heterothermic - enabling them to maintain internal temperatures to some extent (Brill et al. 2001), the limits to their distribution are linked to the limits of their thermal ranges. Temperature is one of the best predictors for tuna species overlap worldwide, and the number of species has been shown to peak at around 20ºC, considerably decreasing between 25–30ºC (Boyce et al. 2008). This is in accordance with the overlap results which show a considerable decrease in the number of species in the areas closer to the equator, areas where sea temperatures are predicted to keep rising above the limits of most species' tolerance (Hastings et al. 2020; Lawman et al. 2022). It is likely that these regions will feature a decrease in the occurrence of the studied tuna, with slow poleward shifts for all species considered. Indeed, previous studies suggested the movement of these tuna species into higher latitudes. Examples of this include a potential shift of *K. pelamis* from more tropical into more temperate waters along the Mozambique Channel (Nataniel et al. 2022) and increasing suitability along the Southern Atlantic Ocean (in the waters of Ascension Island and Saint Helena) for *T. obesus*, *T. alalunga*, *T. albacares*, and for *K. pelamis* (Townhill et al. 2021). At the same time, there are already records of considerable impacts from changing environmental conditions on tuna distribution (Erauskin-Extramiana et al. 2019) and phenology (Chust et al. 2019), and significant changes in tuna biomass are also to be expected (Dueri et al. 2016; Erauskin-Extramiana et al. 2023).

Within this context, there is a looming risk that fishing communities in the tropics could suffer a significant reduction in tuna catches, with the resulting socio-economic impacts increasing along with each community's level of dependency on these fisheries (Sumaila et al. 2016). It is anticipated that the impact's magnitude will decrease as the mobility of the fishing fleet increases, meaning that more technologically advanced fleets are more likely to be better prepared and move to other fishing grounds or change gear. On the other hand, domestic fleets have less ability to adjust because they are confined to their Economic Exclusive Zone (EEZ) (Sumaila et al. 2011). Pelagic tuna species track suitable habitats for feeding and spawning migrations (Lan et al. 2018). Tunas also adopt a nomadic strategy in small schools when food resources are scarce and assemble in larger schools when food resources are abundant suggesting that large aggregations are generally located in the preferred mixed layer of selected locations during migratory movements (Lopez et al. 2017). Those movements to different areas can potentially disrupt their breeding and feeding grounds (Vaihola and Kininmonth 2023). Thus, being the habitat suitability impacted, the preferred locations could also be impacted and, consequently, fishing activity will need to adjust. The need for finding alternative fisheries for displaced fishers and fishing capacity adjustment due to climate change had already been foreseen (McIlgorm et al. 2010).

It could also be important to consider the impact of fish aggregating devices (FADs), which have been defined by ICCAT as “Permanent, semi-permanent or temporary object, structure or device of any material, man-made or natural, which is deployed and/or tracked, and used to aggregate fish for subsequent capture”. FADs can either be anchored (aFADs) or drifting (dFADs) (ICCAT Recommendation 19−02, 2019). They have been intensely used (especially aFADs) in tropical tuna fisheries to increase industrial purse-seine efficiency (Jaquemet et al. 2011; Dupaix et al. 2024), taking advantage of tunas’ natural attraction to floating objects in the open ocean, whether natural or man-made (ISSF and IPNLF 2019). The massive use of FADs has raised concerns about the sustainability of this fishing method (e.g., Jaquemet et al. 2011; Dagorn et al. 2013; Nooteboom et al. 2023) and studies indicate that the impact of FADs can be location-, species- and age-dependent (Dupaix et al. 2024). Although a recent review suggested a lack of clear converging scientific results on the impact of dFADs on tuna movements and life-history parameters at the population level (Dupaix et al. 2024), certain hypotheses have been proposed. For instance, the ecological trap hypothesis (Marsac et al., 2001) suggests that FADs migh attract tunas to areas that are ecological suboptimal—where environmental conditions may not support their survival or reproduction in the long term. That raises important questions in the context of climate change. As oceanographic conditions shift, the combination of anthropogenic structures such as FADs and changing habits could exacerbate mismatches between tuna behaviour and environmental suitability. This interplay between FAD-induced aggregation and climate-driven habitat changes deserves closer attention, particularly for effective spatial management and conservation of tropical tuna populations under climate change scenarios (Maufroy et al. 2017; Dupaix et al. 2024).It is important to discuss the potential limitations of the present study’s results, mainly linked to the availability of data and to the SDM methodology and assumptions (Araújo and Guisan 2006; Heikkinen et al. 2006; Fitzpatrick and Hargrove 2009; Shipley et al. 2022). SDMs are commonly associated with the problem of overprediction, with models predicting a given species to occupy all the suitable areas in the available spatial extent, ignoring real-world limitations such as other unaccounted predictors, geographical or temporal barriers to dispersion, among others (Araújo et al. 2005). Thus, knowledge on species’ ecology and a higher focus on overall tendencies of habitat suitability change are important, as they move models away from specificities of the results which might be related to overpredicting. At the same time, SDM results can also feature a degree of underprediction, which is usually linked to the nature of the occurrence data (i.e., namely regarding sample size and extent). This is highly influenced by biases in spatial sampling and is common in occurrence data from single studies or from online repositories such as GBIF. Indeed, these repositories are unlikely to represent a complete sample of a species’ real-world distribution, as it is likely that these databases suffer from sampling effort, which is geographically skewed (Beck et al. 2013). Another consideration towards discussing potential biases associated with occurrence data, is the use of fishery-dependent data. Indeed, this type of data often are collected by scientific observers on commercial fishing vessels or from logbooks. This data, while advantageous due to the fact that large amounts of data can be gathered, is inherently non-probabilistic, as fisheries either follow fish stocks or fish what’s available during fishing seasons (Karp et al. 2023)—which ultimately leads to data being inherentely biased both spatially and temporally, since the locations of fishing activity are dependent on factors such as species abundance (Conn et al. 2017; Pennino et al. 2019). To overcome this issue, it is important to complement data from different sources, even when dealing with online data repositories. In the case of the present study, this was achieved by sourcing data from the ICCAT, which holds large datasets of several Atlantic-wide oceanographic surveys with a focus on tuna species—fishery-independent data. The use of both types of data has been shown to mitigate (to some extent) the sampling bias associated with the use of only fishery-dependent data, supplementing the available data and resulting in SDMs with high predictive capability (Karp et al. 2023).

In regard to sample size, all species featured sufficient occurrence data to ensure predictive accuracy—according to Stockwell and Peterson (2002), sample sizes under 100 occurrences can make models prone to decreased predictive accuracy, depending on the chosen algorithm. In the case of the present study, the minimum sample size available was just under 600 points after environmental sub-sampling (i.e., for *T. thynnus*; see Table 2 in the methods section). Accordingly, the 'megaSDM' package kept all model runs for each species during model evaluation procedures, which suggests their performance to be reliable and their results able to be interpreted. SDMs are considered a very effective methodology in addressing some of the questions regarding the climate change effects on biodiversity, such as changes to biogeography, species hotspots and climate-refugia (Sinclair et al. 2010). Likewise, the overall tendencies displayed by the different species reflect the major climatic drivers of change and are then bound to be meaningful (Garcia-Callejas and Araujo 2016). Future studies should aim to include the potential impact of changing environmental conditions on fishery intensity, as it will contribute to shaping the long-term distribution of these tuna species, by introducing further anthropogenic pressure (Barange et al. 2010; Schick and Lutcavage 2009). Lastly, while MaxEnt is considered to be a good choice for single-algorithm SDM, future studies could benefit from testing different algorithms and their respective ensembling as this can help to build upon upon the confidence in the predictive performance and stability of the models.

These changing conditions, driven largely by climate change, will continue to test the ability of science and management bodies to adapt to change (Lomonico et al. 2021). Considering the economic and socioecological value of tuna species, predicting responses to climate change could assist the design of more effective management to ensure not only the sustainability of tuna populations but also the sustainability of human communities dependent on them (Erauskin-Extramiana et al. 2019). The design of management in response to climate change must be designed specifically for the species, considering the different gears used and suited for the area being managed (Mediodia et al. 2020). Furthermore, anticipatory management measures rather than reactive ones are likely to have better control over negative economic impacts (Sumaila et al. 2011). Principles for climate-adaptive fisheries management have been discussed by Free et al. (2020), including the implementation of better practices in fisheries management, namely a dynamic, flexible, and forward-looking management, encouraging international cooperation and building economic and socioecological resilience. Effective partnerships among management agencies, fishing industries, the private sector and academia could fill the capacity gaps to achieve climate-ready fisheries (Lomonico et al. 2021).

Final remarks

Environmental and social dimensions of fisheries are linked by climate change. The projection of climate change impacts on the distribution of tuna population and their inclusion in collaborative approaches enables the adequate and early planning of potential mitigation measures, both economic and socioecological.

The present study contributes to the body of research on the biogeographical impacts of climate change on the five tuna species most captured in the Atlantic Ocean, with the aim of contributing towards fisheries management, by providing a set of future projections where the identification of potential areas of increase/decrease in habitat suitability and the consequent potential changes in each species distribution towards the end of the century are possible. By identifying such areas, it is possible to complement potential stock evaluation and management, and alert on the need for better practices in fisheries management and achieve climate-ready fisheries, dynamically building resilience.

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Author contributions PMS designed the research topic. FOB performed the statistical analyses. PMS and FOB wrote the manuscript. CMT, CP and JLC reviewed the manuscript.

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Data availability Data is provided within the manuscript or supplementary information files.

Declarations

Competing interests The authors declare no competing interests.

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